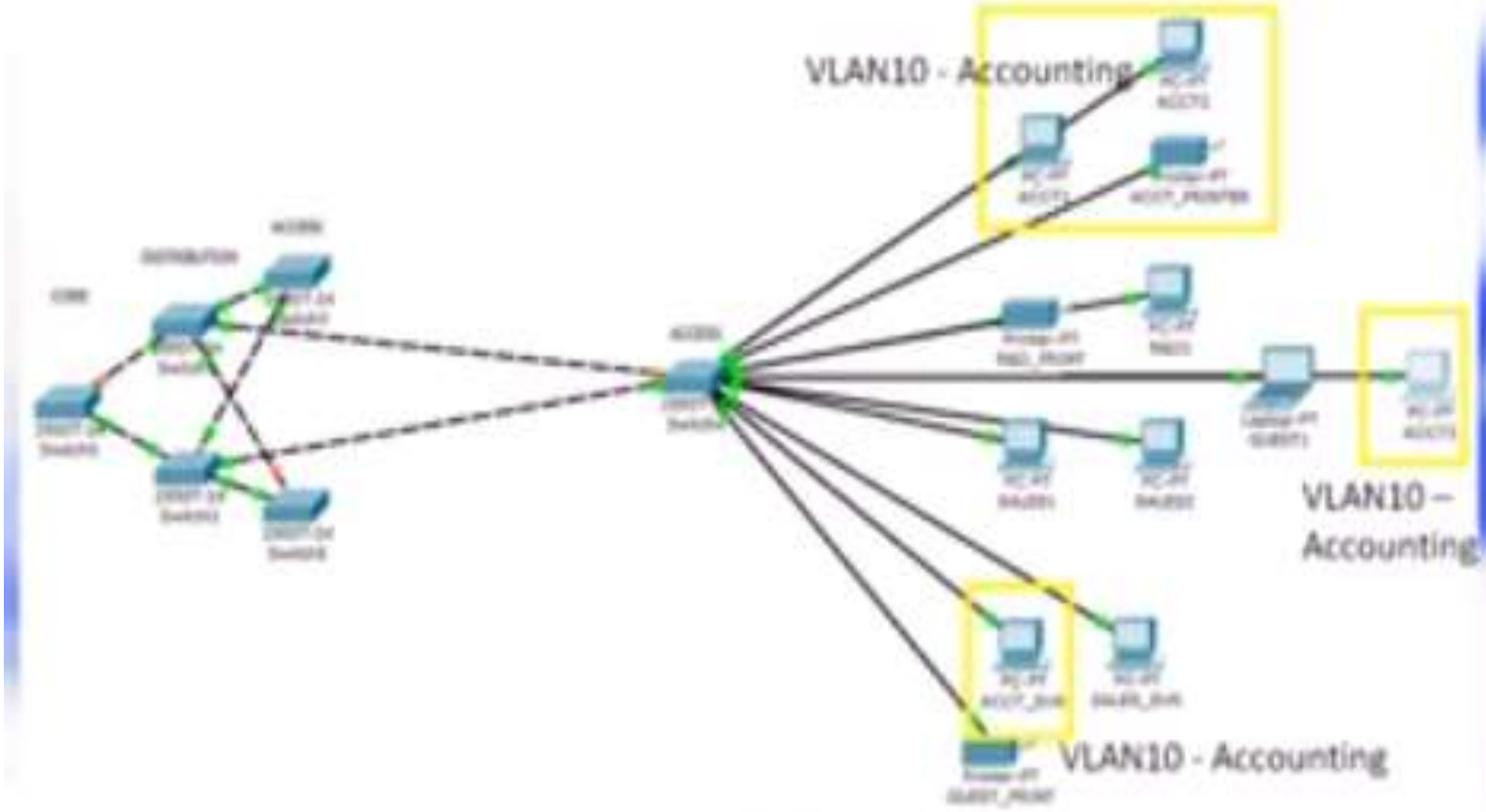
What are VLANs?

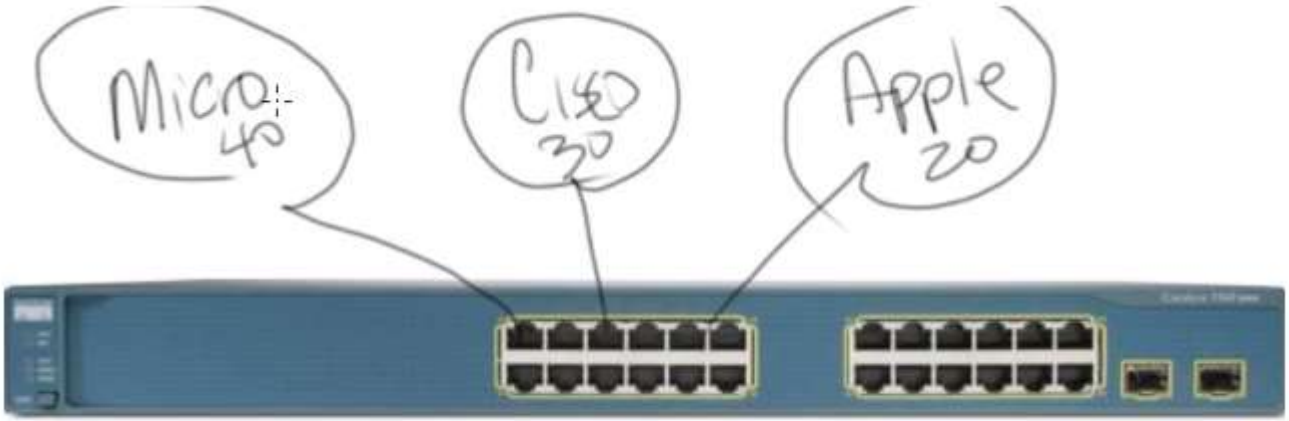
- VLAN stands for Virtual Local Area Network (Virtual LAN)
- A VLAN is a group of logically network devices.
  - Such as a set of networked computers and printers for a department or building floor.
  - Can separate network "guests" and trusted users traffic.
- A logically separate subnetwork
  - Devices on VLAN 20 and VLAN 30 can't communicate without a layer 3 device.

What can they do?

- Create smaller broadcast domains, and therefore less wasted bandwidth.3
- Increase security, as VLANs are not visible to outside traffic
- Decrease costs
  - Buildings with multiple companies can use a single network infrastructure.
  - Efficient use of bandwidth (2 trunks for a high traffic VLAN)
- Simplify management

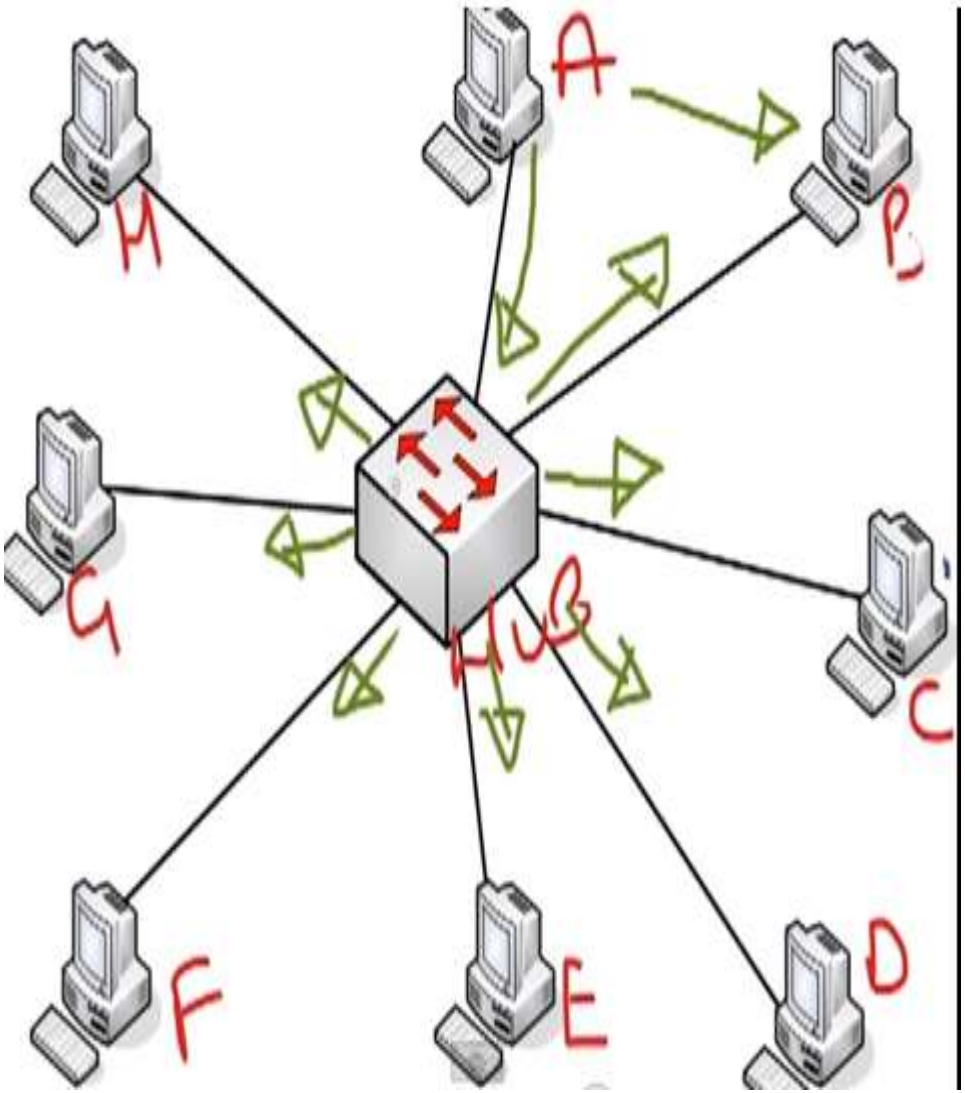
VLANs Illustrated



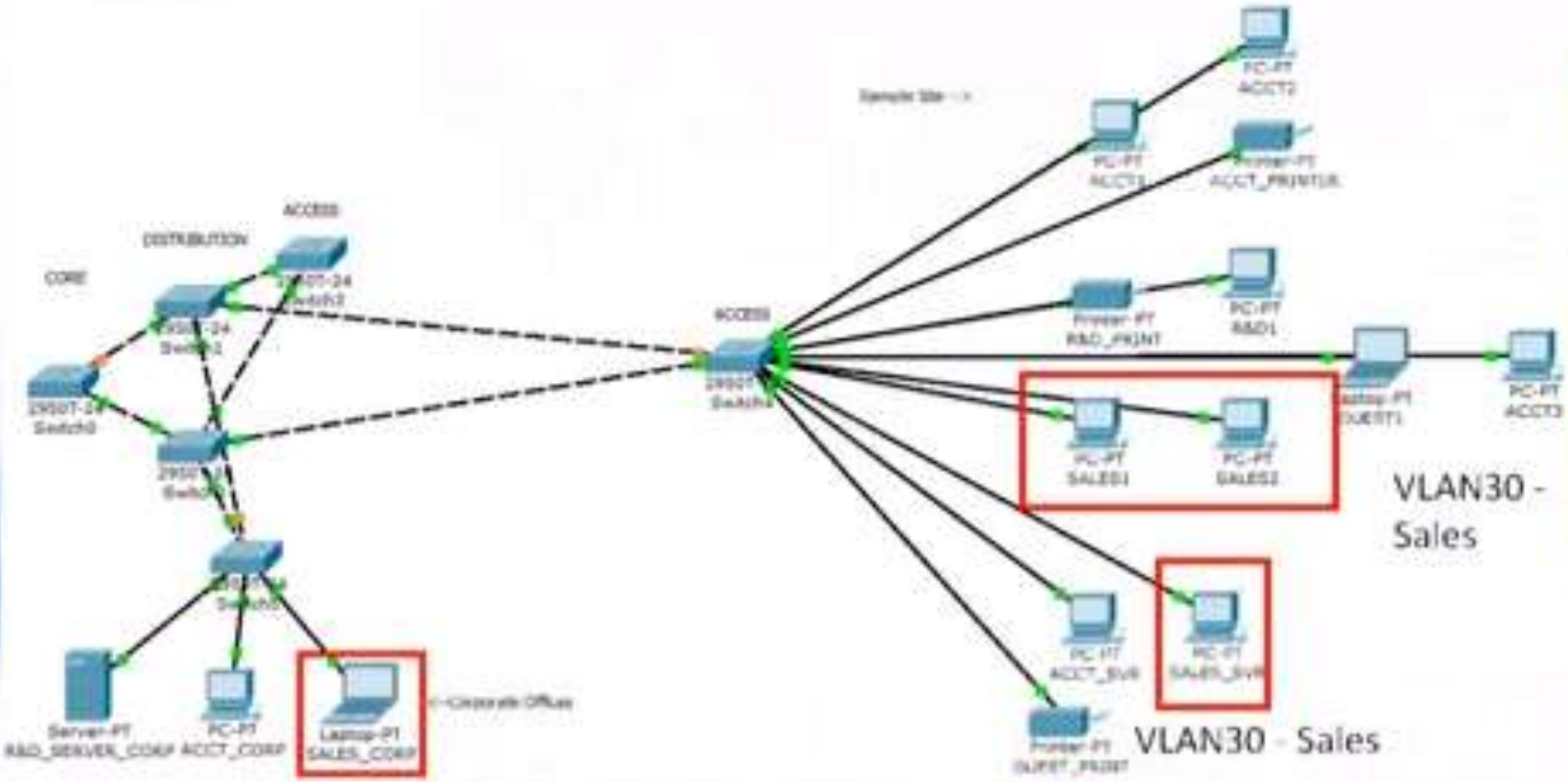

Micro
40
Cisco
30
Apple
20

Virtual LAN. Group of devices on one or more LANs that are configured (using management software) so that they can communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments.

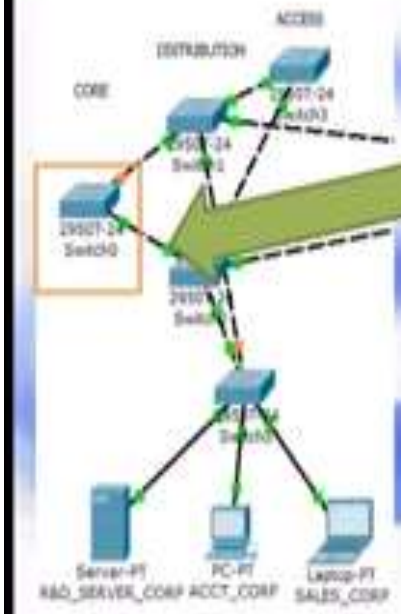
Because VLANs are based on logical instead of physical connections, they are extremely flexible.


A
B
C
D
E
F
G
H
HUB

VLANs Illustrated



VLANs – Adding to Network



- Using the CLI, we enter the following on switch CORE1

```
CORE1(config)#vlan 10
CORE1(config-vlan)#name accounting
CORE1(config-vlan)#vlan 20
CORE1(config-vlan)#name R&D
CORE1(config-vlan)#vlan 30
CORE1(config-vlan)#name sales
CORE1(config-vlan)#vlan 13
CORE1(config-vlan)#name guest
CORE1(config-vlan)#exit
CORE1(config)#
```

- The VLANs are then distributed to all of the switches via VTP.

VLANs – Configuring Ports

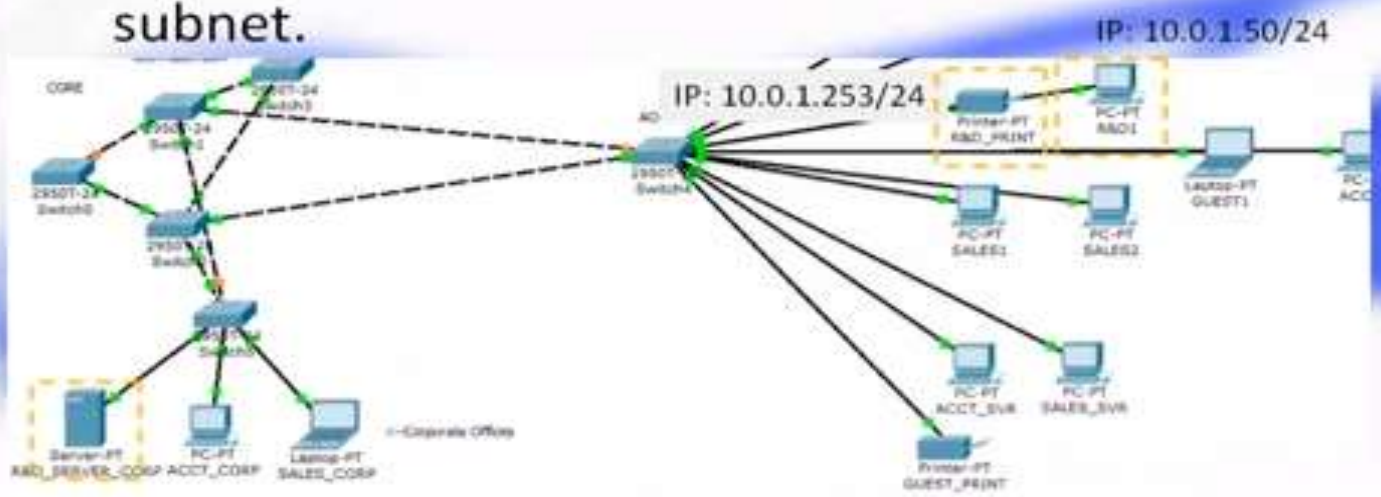
- On each switch, identify which device is supposed to be on which VLAN. For example, R&D_SERVER_CORP needs to be on VLAN 20. It is connected to Gigabit Interface 2.

```
CORP_ACCESS(config)#int gi1/2
CORP_ACCESS(config-if)#switchport
mode access
CORP_ACCESS(config-if)#switchport
access vlan 20
CORP_ACCESS(config-if)#exit
CORP_ACCESS(config)#
```

- Do the same on all switches, setting the correct ports to the correct VLAN.

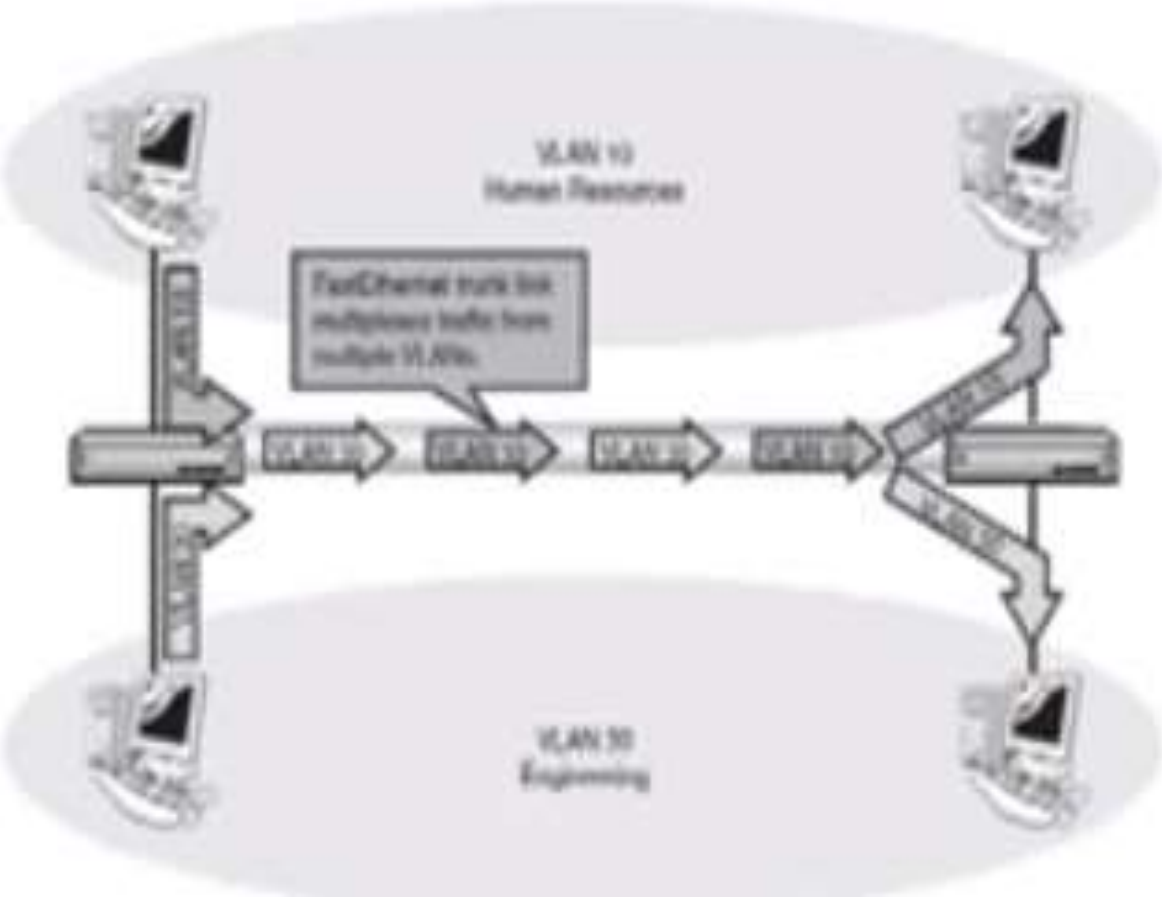
VLANs – Configuring Devices

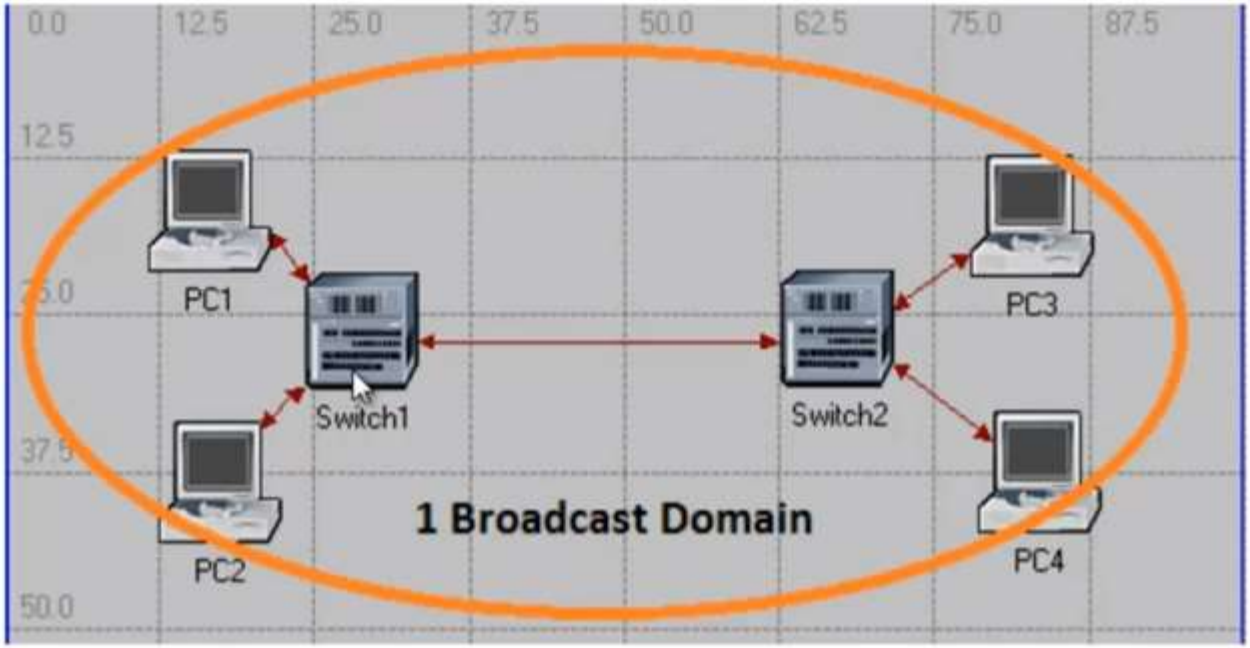
- On the device end, the only note is that all devices on a VLAN must be on the same subnet.



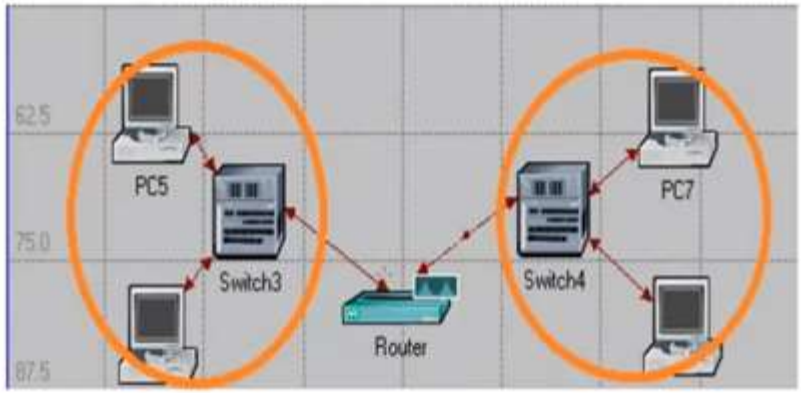
IP: 10.0.1.254/24

Trunking


VLAN 10
Human Resources
FastEthernet trunk link
multiplexes traffic from
multiple VLANs.
VLAN 50
Engineering

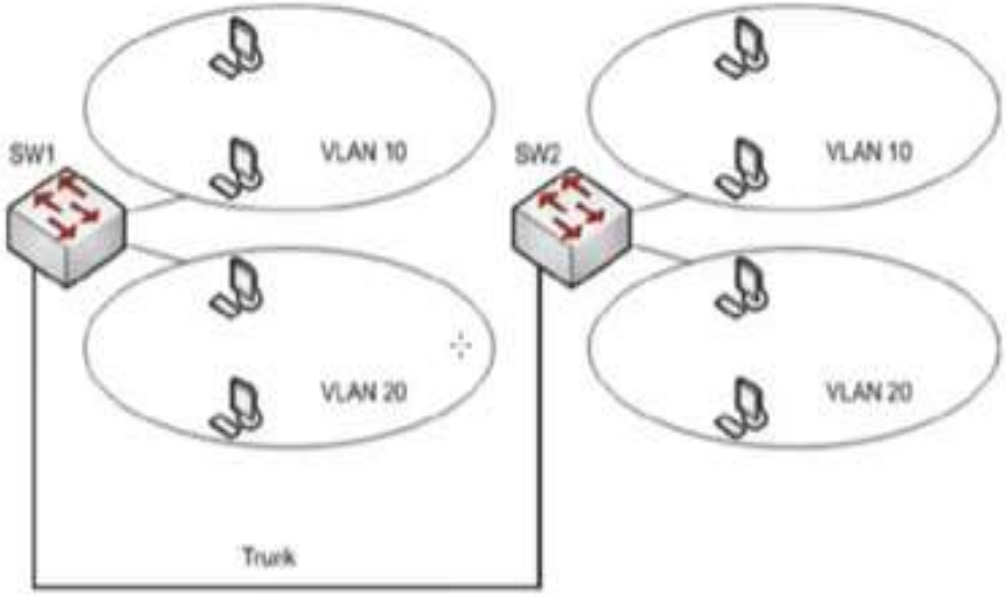

0.0
12.5
25.0
37.5
50.0
62.5
75.0
87.5
12.5
25.0
37.5
50.0
PC1
PC2
PC3
PC4
Switch1
Switch2
1 Broadcast Domain

- Two broadcast domain
- Router acts as the boundary



Virtual LAN

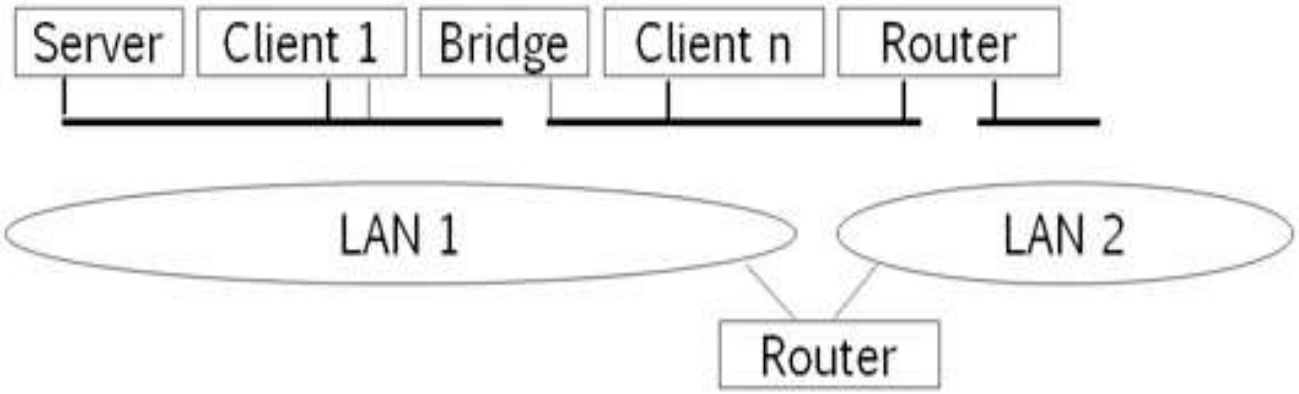
logically separates the traffic flow in a network regardless of location



Laboratory Scenario

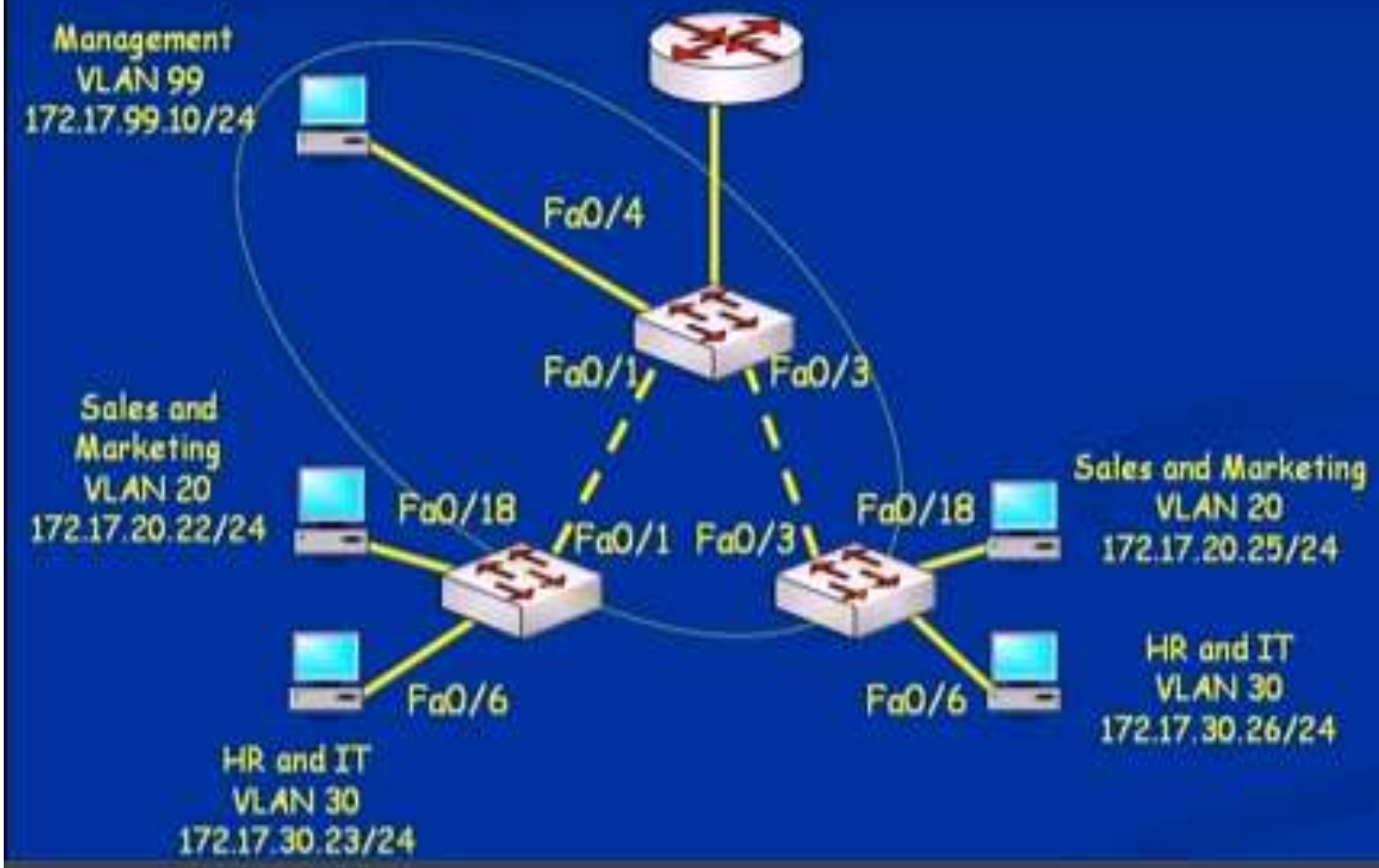
ICareMore company had acquired a building beside its main building. The company would like to setup 5 workstations in each of the three department: Billing, Sales, Tech. However, this time the company requires to separate the application traffic in each department because the Billing department is accessing sensitive information. A server in each department will be provided and will be located in main building's second floor. Let us find out what happens when we separate the traffic using Virtual LAN.

What is a LAN?



› LAN = Single broadcast domain = Subnet

› No routing between members of a LAN

› Routing required between LANs

- What is a VLAN and why needed?
- VLAN = a virtual division of the physical Switches [creating logical ports]
- Reasons = To separate traffics from each other.
- To create different subnets.
- To restrict access to the network and provide better security.